Trayecto 1 -> Exponencial

$x = [0 \text{ a } 8]$ $\quad x(t) = t \quad \dot{x}(t) = 1$

$f(x) = e^{0.2t} \quad y(t) = e^{0.2t} \quad \dot{y}(t) = 0.2\, e^{0.2t}$

Condiciones iniciales

$x(0) = 0$

$y(0) = e^{0.2(0)} = 1$

Velocidad → $v(t) = \sqrt{(1)^2 + (0.2e^{0.2t})^2}$

Orientación → $\theta(0) = \tan^{-1}\left(\frac{\dot{y}}{\dot{x}}\right) = \tan^{-1}\left(\frac{0.2\, e^{0.2(0)}}{1}\right) = \tan^{-1}(0.2)$

→ $w(t) = \frac{d(\tan^{-1}(0.2\, e^{0.2t}))}{dt} = \frac{e^{0.2t}}{e^{0.4t} + 25}$

Velocidad angular (derivar posición θ)

TRAYECTORIA 2:

Cosenoidal

$x = [8 \text{ a } 18]$ $\quad x(t) = t \quad \dot{x} = 1$

$f(x) = 3\cos(0.4(t-8)) \quad y(t) = 3\cos(0.4(t-8)) \quad \dot{y} = -1.2\sin(0.4t - 3.2)$

$\ddot{y} = -0.48\cos(0.4t - 3.2)$

$v t = \sqrt{\dot{x}^2 + \dot{y}^2}$

Condiciones Iniciales

$x(8) = 8$

$y(0) = 3\cos(0.4(8-8))$

$\theta(t) = \tan^{-1}\left(\frac{-1.2\sin(0.4t - 3.2)}{1}\right)$

$\theta(0) = \tan^{-1}(-1.2\sin(0.4(8) - 3.2))$

↓ Derivada

$w(t) = -\frac{0.48\cos(0.4t - 3.2)}{1 + 1.44\sin^2(0.4t - 3.2)}$

TRAYECTORIA 3

$x = [18 \ a \ 30]$

$f(x) = 2\ln(t-17)$

$x(t) = t$

$y(t) = 2\ln(t-17)$

$\dot{x}(t) = 1$

$\dot{y}(t) = \frac{2}{t-17}$

$\ddot{y}(t) = \frac{-2}{(t-17)^2}$

$\theta(t) = \tan^{-1}\left(\frac{2}{t-17}\right)$

$w(t) = -\frac{2}{t^2-34t+293}$

Condiciones iniciales

$x(18) = 18$

$y(18) = 2\ln[18-17]$

$\theta(18) = \tan^{-1}\left(\frac{2}{18-17}\right)$